

Detailed Module Description

1. Admin Module

The Admin Module serves as the system's control panel, allowing administrators to manage participants and ensure secure assessments.

Key Features:

- User Management: Add, update, or remove students, applicants, and interviewers.
- Scheduling: Organize exams/interviews and generate secure, encrypted participation links.
- AI Configuration: Define proctoring rules such as multiple face detection, prohibited object alerts, and eye movement tracking.
- Reports & Logs: Access AI-generated proctoring reports, incident logs, and system analytics.

2. User Module (Candidate/Applicant)

This module provides the candidate interface for joining and attending exams or interviews.

Key Features:

- Login & Identity Verification: Secure authentication using face recognition and ID proof.
- Secure Session Access: Join scheduled assessments through encrypted one-click links.
- Live Proctoring: Continuous monitoring via webcam and microphone.
- Real-Time Alerts: Notifications such as "Stay in frame" or "No background noise".
- Submission: Upload answers for tests or participate in live interview sessions through a protected interface.

3. Interviewer/Examiner Module

The Interviewer Module provides tools for evaluating candidates and managing assessments.

Key Features:

- Question & Script Creation: Prepare question sets or interview guidelines.
- Candidate Monitoring: Supervise candidates live or review flagged recordings later.
- Report Access: View AI-generated analytics reports, including flagged events and compliance scores.
- Evaluation & Feedback: Assign marks, record observations, and provide feedback securely.

4. AI Proctoring Engine (Core Backend Module)

The AI Proctoring Engine is the backbone of the system, ensuring fairness and transparency during assessments.

Key Features:

- Face Detection & Recognition: Verify that the correct candidate is present.
- Object Detection: Identify unauthorized materials like books, phones, or extra people.
- Voice Analysis: Detect background voices, noise, or abnormal candidate speech.
- Eye & Head Tracking: Identify suspicious activities such as looking away frequently.
- Alerts & Logging: Automatically generate alerts for rule violations and store them in logs.
- AI Reports: Compile behavior data and generate detailed proctoring reports for admins and interviewers.